Aidan Schreder
Systems Design Engineer at the University of Waterloo

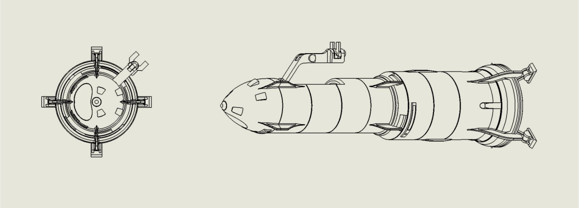
Mechatronics Portfolio

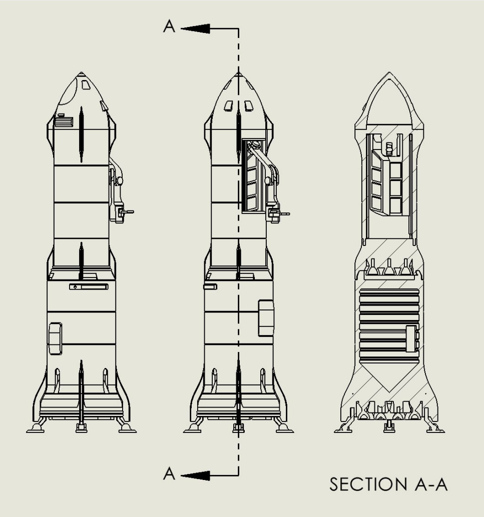
Selected extracurricular projects emphasizing mechatronics, fabrication, and iterative testing under real-world constraints.

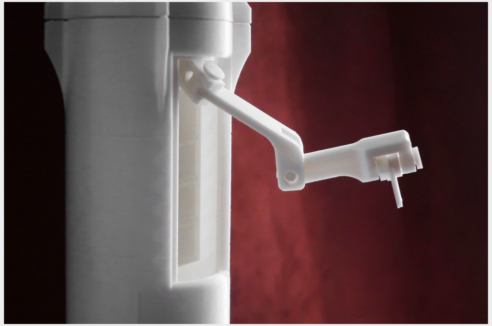
Satellite Repair Ship CAD Model and 3D Print

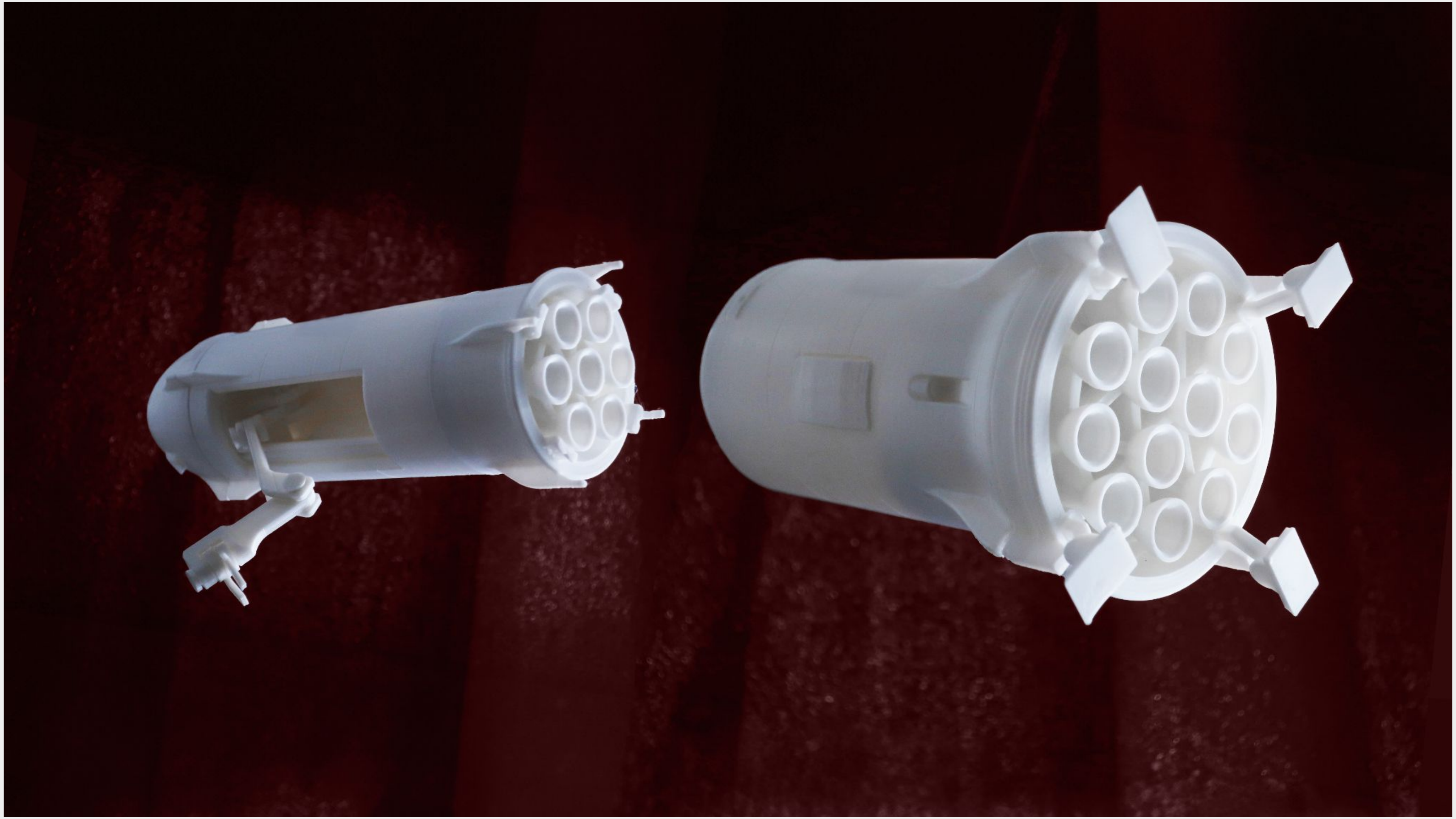
I designed and 3D printed a scaled spacecraft of my own design: a **two-stage** satellite repair vehicle with a **fully articulated, printable robotic arm**, a deployable door, and mechanical stage locking, assembled without adhesives.

Early prototypes failed due to joint binding and structural fragility, which I addressed through multiple redesign-and-test iterations involving adjusted tolerances, wall thicknesses, print orientations, and variable infill to achieve reliable motion.









SOLIDWORKS, Creality White PLA, January 22, 2025

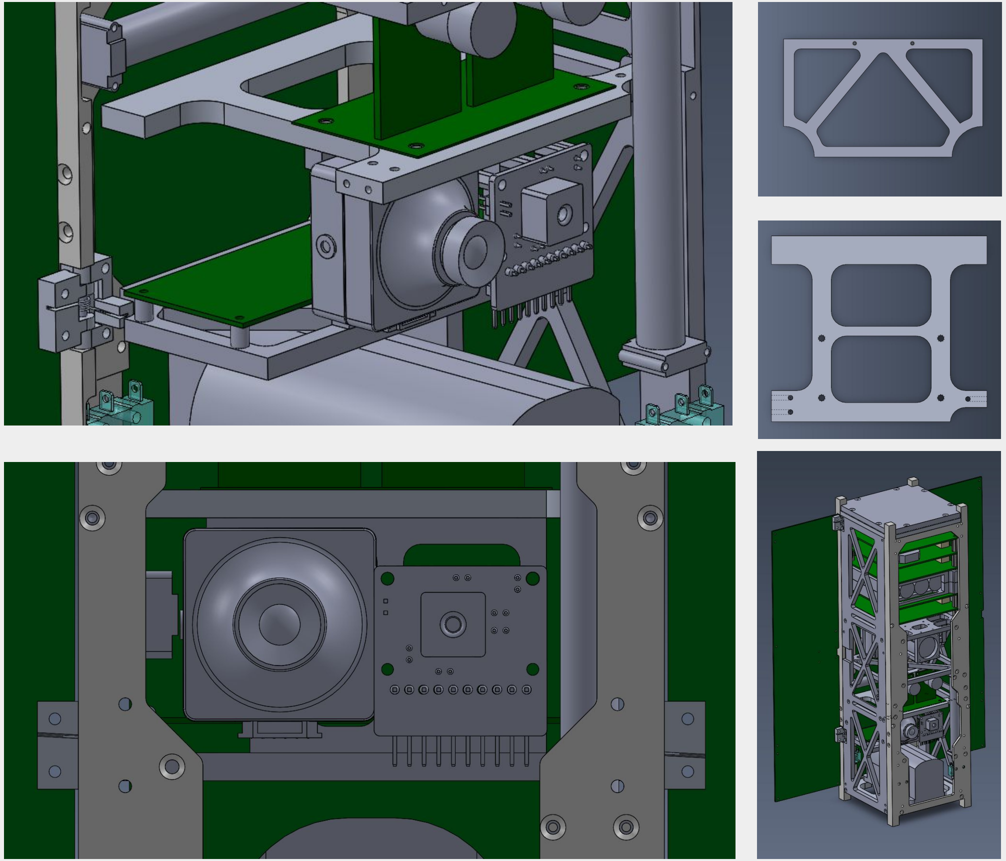
IR & Visible Camera Satellite Interface

Mechanical Research and Design

I am currently designing the mounting interface for a dual-camera imaging payload for CubeSat deployment. It integrates a Lepton 3.1R thermal IR sensor and an Arducam visible-light sensor, to contribute to the project's goal of **detecting Resident Space Objects (RSO's).** In order to select the cameras, I evaluated **resolution, size, and performance trade-offs** of several modules.

I also compiled detailed studies of both cameras that summarizes mechanical, electrical, and software integration details. Both the Arducam Study and Lepton Study are available online here.

I have iterated on the mount design several times to fit within the **90 x 94 mm frame** and accommodate wiring and PCB constraints as well as **vibration-isolation requirements.**



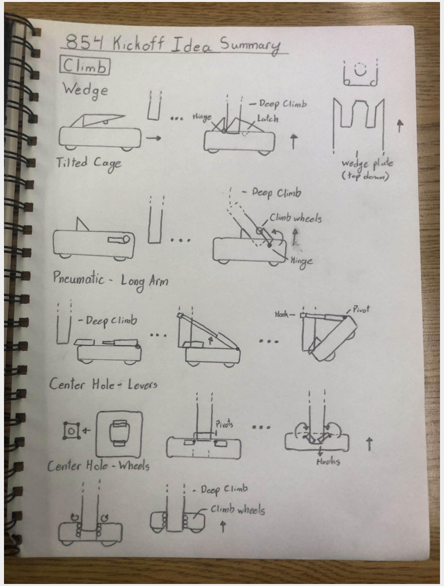
Solidworks, Aluminum Frame, May 20, 2026

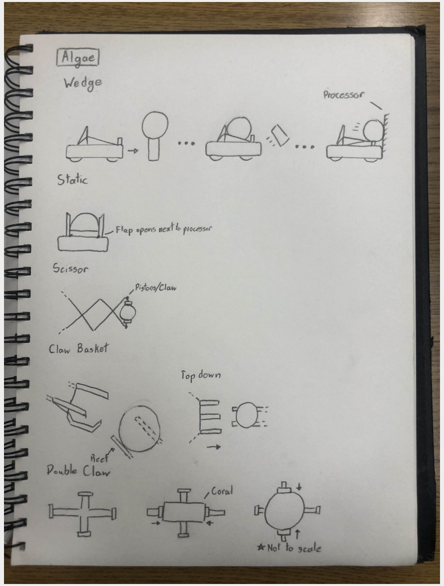
FRC Reefscape Robot

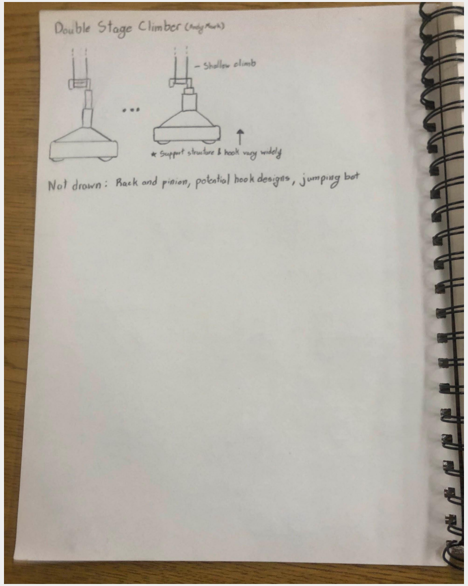
Rapid Prototyping, 3D Design, Manufacturing

As the Captain of FIRST Robotics Competition (FRC) Team 854, I was tasked with directing the **development of a human sized robot** for the 2025 FRC Reefscape season. All following images were created or taken by me.

Development began with brainstorming at our team's kickoff event, where general members split into groups and pitched their ideas to our exec team. I documented ideas, added my advice, and curated them into an idea summary in my notebook.



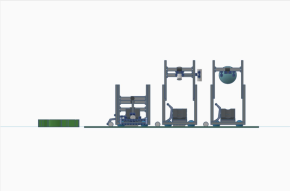


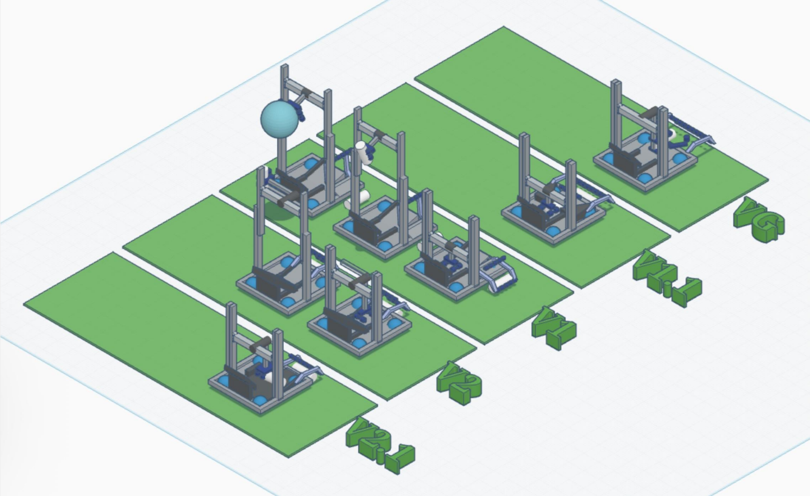


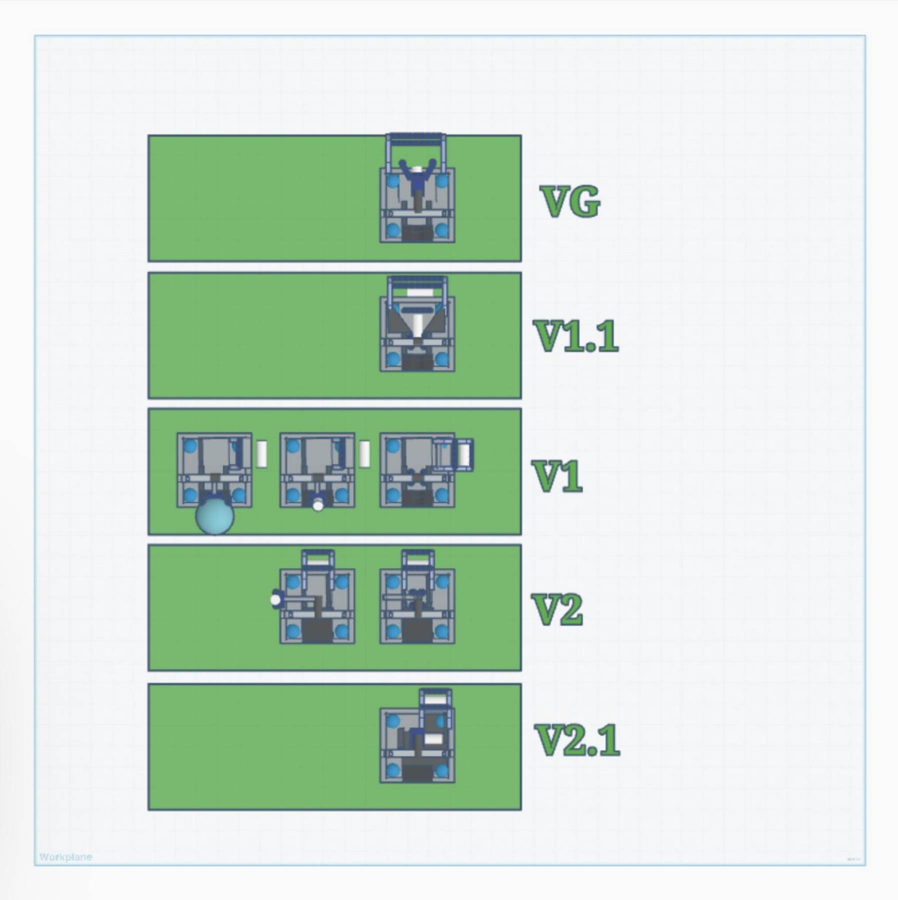
FRC Reefscape Robot

Rapid Prototyping, 3D Design, Manufacturing

After rigorous team discussion, we decided which designs had the most potential. I blocked out these ideas in 3D to validate dimensions, experiment with orientation, and provide a visual guide for others to reference.







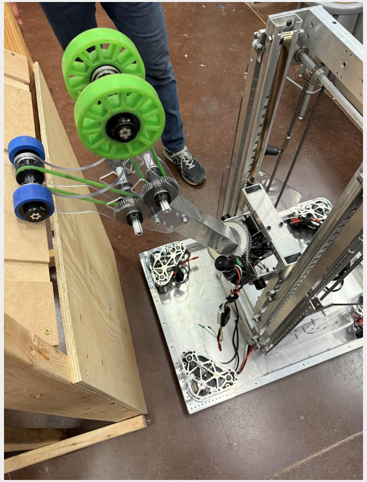
FRC Reefscape Robot

Rapid Prototyping, 3D Design, Manufacturing

In the manufacturing and prototyping stage, I helped create wood replicas with metal axes of planned intake mechanisms to **test game-piece compression and efficiency**. Our final intake design and robot are pictured on the right. Key challenges included **cable routing reliability, drivetrain integration tolerances, and rapid iteration** under competition deadlines. I directly supported electrical routing strategies to reduce snagging and fatigue, and assisted in mechanical revisions after early testing revealed steering instability.



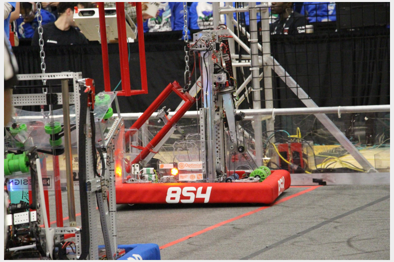


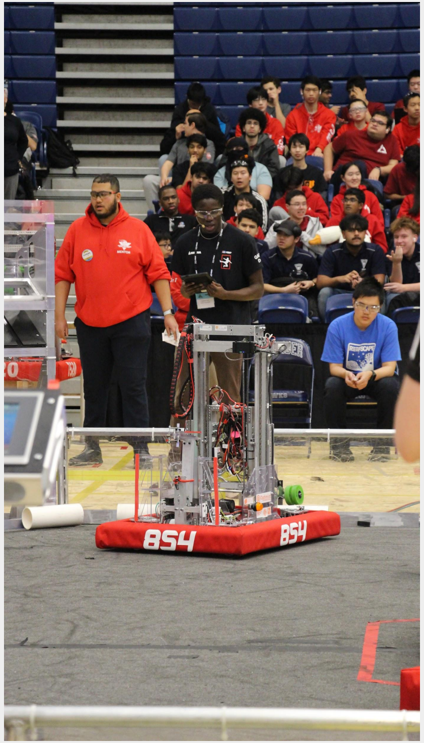


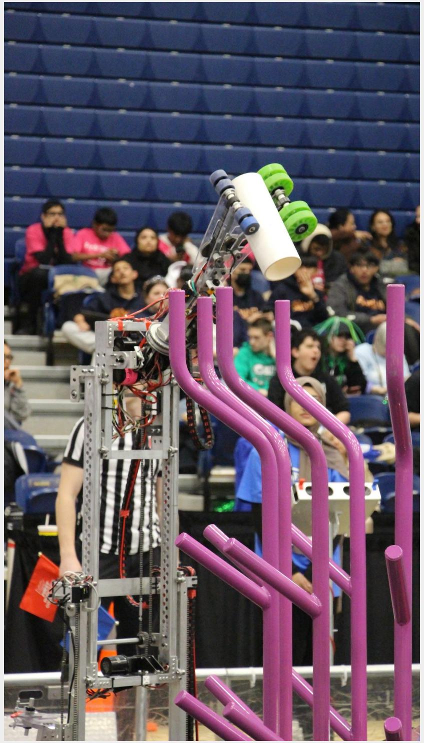
FRC Reefscape Robot

Rapid Prototyping, 3D Design, Manufacturing

After we completed the design, I riveted, machined, and assembled components to achieve our final robot design. One of our toughest challenges was cable management - I assisted our electrical leads with routing strategies. Pictured to the right is our robot at competition, including competing at the provincial level.







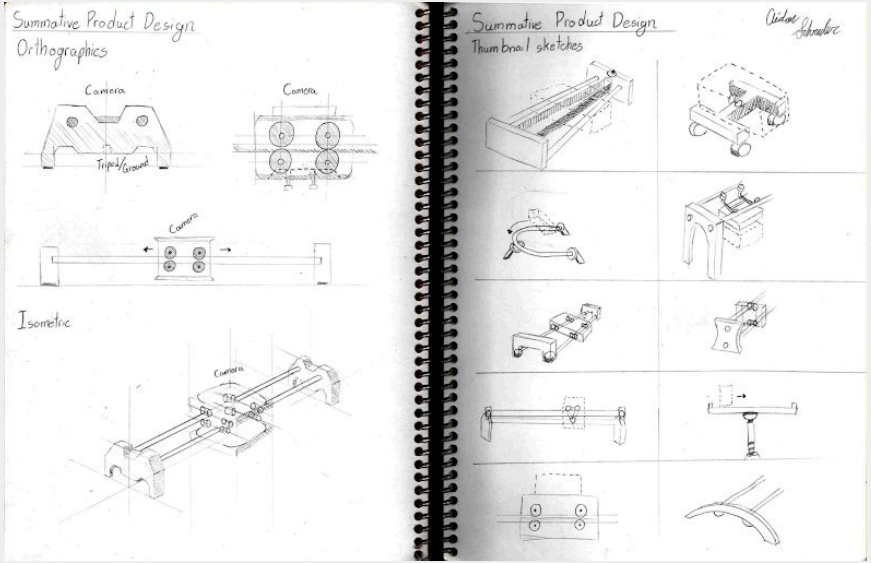
Camera Roller

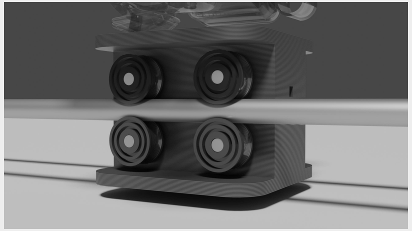
Product Design

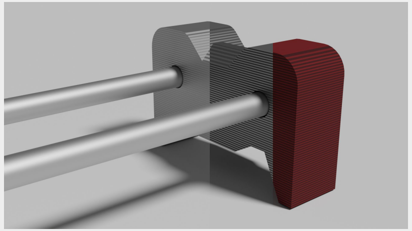
My inspiration for this project stemmed from my **love of videography** and the **expensive nature of film equipment**. Entry-level accessories can easily cost up to $1000. To help with videography projects and reduce equipment costs, I designed and fabricated a stabilizer/dolly system using parts I 3D printed and readily available hardware store parts. I have used the Camera Roller in several film projects since.

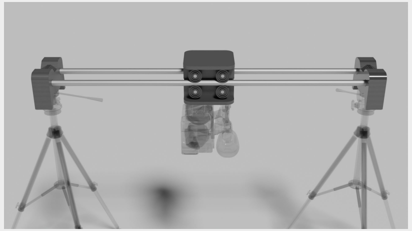
The stabilizer is easy to use (branching into my goal of making filmmaking accessible); a camera is placed on the rolling platform and can be slid smoothly across the rails to create a side-scrolling shot. Key constraints included minimizing vibration, and achieving smooth motion using low-cost hardware-store components.

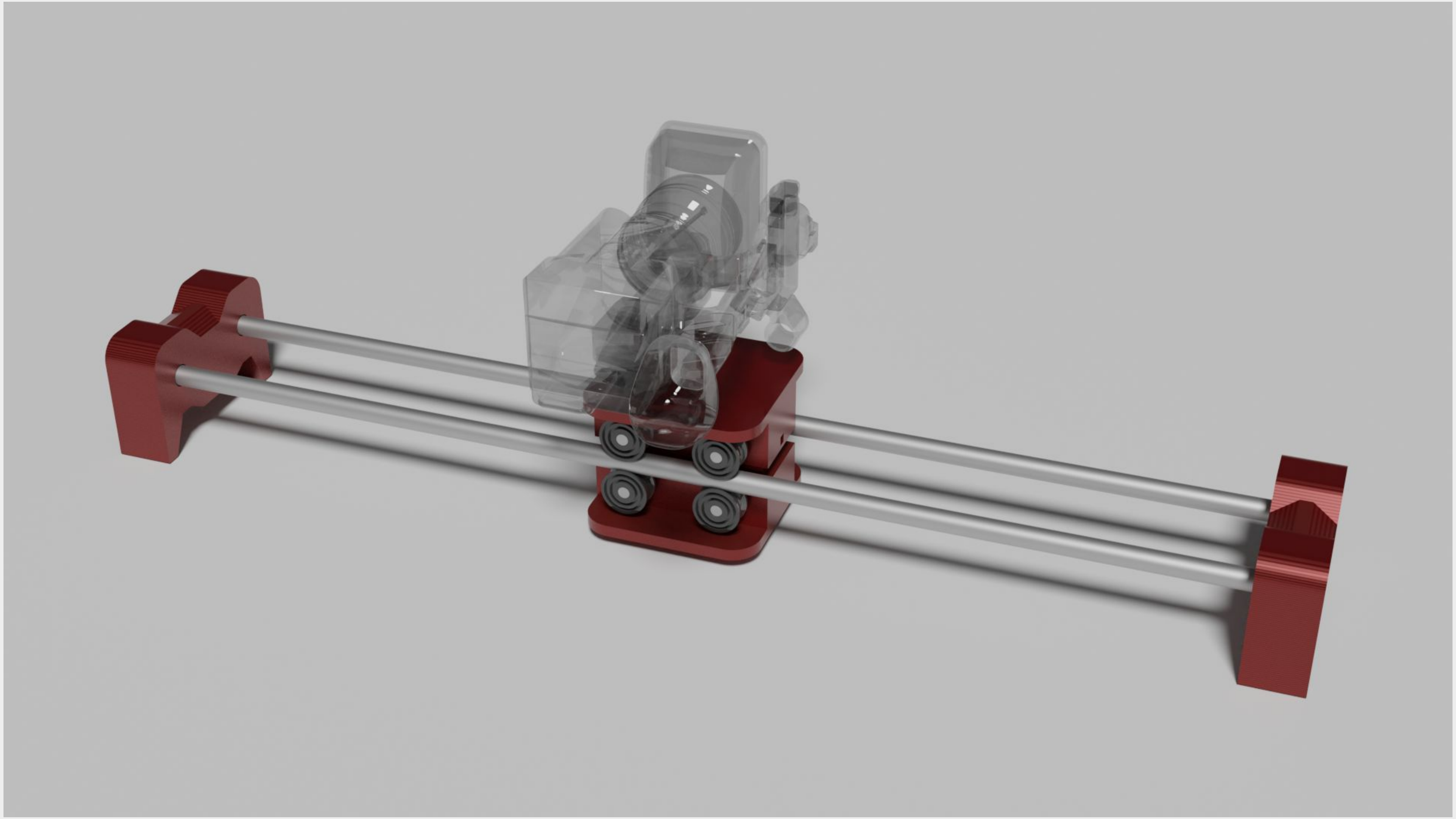
[Read a full report on my Camera Roller here.]











Fusion 360, Adobe Photoshop, PLA Plastic, Aluminum, January 16, 2024, 40” X 7” X 4.5”

Thank you for viewing!

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